

Kyle Tran

MECHANICAL ENGINEERING • ROBOTICS MINOR • NORTHEASTERN '29

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Seeking a Summer 2027 co-op in hardware and robotics. I build robots, autonomous vehicles, and embedded systems — drawn in CAD, machined and printed, wired and coded, then tested until they hold up: two years of competitive combat robotics, and an autonomous survey boat built from scratch.

4th NATIONALLY, 2024	3.87 GPA	10 ROBOTS BUILT	16 TEAM PODIUMS
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D.R.E.A.M.S — autonomous survey boat



Combat robots — Team DHM



Maze-navigating robot

PROJECTS

Autonomous Environmental Survey Boat — D.R.E.A.M.S

Jan 2026 — Present

A twin-hull catamaran that surveys lakes on its own. v1, built and field-tested, completed autonomous waypoint surveys on local lakes, winning an instrumented pod to the bottom to sample the water. v2, in progress, adds a Raspberry Pi 5 with stereo-camera obstacle avoidance, a custom KiCad board in bring-up on its second revision, and 4G telemetry, all validated in simulation first. The hull, electronics, PCB, and integration are my own work; the software was built with AI coding agents.

SolidWorks • ASA + fiberglass hull • KiCad PCB • Pi 5 + 2x RP2040 • MicroPython • YOLOv8 • VESC • GPS/IMU • 4G + PostGIS

Combat Robotics — Team DHM

2023 — 2025

Designed and built around ten robots that competed across five weight classes — weapon and drivetrain design, titanium and steel armor, composite layup, and iteration driven by what broke in the arena. 4th nationally in Plastic Antweight (2024), U.S. rankings of top-15, top-30, and top-35, and 4 firsts and 8 seconds at regional events; the club took 16 podiums over 5 events. The newest Achilles walks on a single-cam shuffler drive I designed.

Onshape • CNC-machined aluminum (outsourced) • laser-cut Ti / steel • brushless drive • FDM • DFM to a weight limit

Adaptive Compliant Robotic Arm

2026 — in design

A five-axis arm with a compliant TPU gripper that reads its own grip force. The four joints below the wrist run 17:1 cycloidal gearboxes with magnetic encoders on the output shaft, each ring sized against pin contact stress and a $\pm 0.25^\circ$ static accuracy budget. Nothing is built yet: the first prototype gets printed to check motion and tune the control loop.

Cycloidal drives • closed-loop steppers • output-side magnetic encoders • TPU gripper • RP2040 • tolerance budgeting

Autonomous Maze-Navigating Robot

Oct — Dec 2025

A 3D-printed tank-drive robot, built with Matthew Hong for the Cornerstone of Engineering course at Northeastern, that completed all three challenge courses. I did the CAD, the printing and the MicroPython firmware: proportional steering off two angled ultrasonic sensors, with asyncio so the LCD and LEDs never stall the motors.

Onshape • 3D printing • MicroPython • asyncio • RP2040 • ultrasonic + IR • I2C LCD

Miscellaneous Builds

2023 — Present

10+ custom ceramics tools designed in CAD, printed, and open-sourced for anyone to print; 3D-printed RC aircraft with carbon-fiber spars; a full sewing machine reverse-engineered in CAD; and printed climbing hardware.

CAD to FDM • published on MakerWorld • github.com/ka1e27

Experience & Skills

EDUCATION

B.S. Mechanical Engineering · Minor in Robotics Sep 2025 – May 2029 (exp.) · GPA 3.87
Northeastern University · Boston, MA
Cornerstone of Engineering I · Statics · Thermodynamics · Materials Science · Probability & Statistics

LEADERSHIP EXPERIENCE

President Oct 2023 – Jun 2025
Combat Robots Club, Burlingame

Ran workshops on CAD, soldering, physics, materials, and 3D printing; handled sponsorships and club finances. The team placed on the podium 16 times – 4 firsts, 7 seconds, 5 thirds – across 5 events.

Ceramic Tools R&D Oct 2023 – May 2025
Burlingame High School · Ceramics Program
Modeled, printed, and published 10+ ceramics tools – trimming tools, ribs, wire guides, stamps – free on MakerWorld.

Secretary · Co-founder Jan 2024 – May 2025
Burlingame Aeronautics Club
Helped found the club and taught the physics of flight – aerodynamics, lift, drag – then put it into practice designing 3D-printed RC aircraft.

Subsystem Lead Sep 2021 – May 2024
Iron Panthers FRC #5026
Designed the spindexter mechanism for the competition robot, working with 10+ team members and 5 professional mentors through design reviews and manufacturing.

SKILLS

DESIGN & CAD
SolidWorks · Onshape · Fusion 360 · mechanism design · DFM

ELECTRONICS & FIRMWARE
KiCad · custom PCB · Raspberry Pi · RP2040 · MicroPython · soldering

SOFTWARE & AI
Python/asyncio · MATLAB · Java · AI coding agents · simulation-first development

FABRICATION
2.5D CNC machining · FDM printing · composite layup (FG/CF) · laser-cut Ti / steel

ROBOTICS & AUTONOMY
Stereo camera rigs · GPS + IMU fusion · VESC drive · 4G telemetry · state machines

LEADERSHIP
Club president · workshops · sponsorships · mentoring

AWARDS

4th	Robot Combat League National Championship	1 lb plastic antweight · 2024
1st	CROS – March Madness	1 lb antweight · 2024
1st	BCRC Bell Brawls IV	3 lb beetleweight · 2024
1st	Spartan Bell Slugfest	1 lb plastic antweight · 2024
1st	NorCal Robotics Expo	1 lb plastic antweight · 2023
2nd	Eight further second-place finishes	ant / beetleweight · 2024